

COM3035 – OPERATING SYSTEMS

MIDTERM EXAM QUESTIONS – 10.11.2025

1. (25%) Consider the following C program:

(a) What will be the output of the program?

(b) Will the output always be the same each time the program is executed?

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("A ");

    pid_t pid;
    pid = fork();

    if (pid == 0) {
        printf("B ");
        execl("/bin/date", "date", NULL);
        printf("C ");
    }
    else {
        printf("D ");
    }
    printf("E ");
    return 0;
}
```

2. (10%) What are the three types of interrupts handled in kernel mode. Explain why are they handled in kernel mode?

3. (30%) Consider the following set of jobs to be scheduled for execution on a single CPU system.

Process	Arrival Time	CPU Burst
P1	0	4
P2	4	6
P3	6	1
P4	8	3

- (a) Draw a Gantt chart showing preemptive SRJF scheduling.
- (b) Calculate the average waiting time and average turnaround time for the preemptive SRJF.
- (c) Assume that the process P1 used the CPU for (6ms, 4ms) in its 2 previous executions. What will be the next CPU burst prediction for P1 if the system uses exponential averaging with equal weights for the most recent past and for the earlier CPU usages.
- (d) What is the time complexity $O(n)$ of the preemptive SRJF algorithm applied?

4. (15%) The following pseudocode shows a simple version of the *Producer–Consumer* problem using a bounded buffer.

PRODUCER

```
while (true) {  
    while (counter == BUFFER SIZE)  
        ; /* do nothing */  
    buffer[in] = next produced;  
    in = (in + 1) % BUFFER SIZE;  
    counter++;  
}
```

CONSUMER

```
while (true) {  
    while (counter == 0)  
        ; /* do nothing */  
    next consumed = buffer[out];  
    out = (out + 1) % BUFFER SIZE;  
    counter--;  
}
```

Identify and explain two problems with this solution.

5. (20%) Write a simple client–server program in Java or C that communicates over TCP sockets. The server should create a socket that listens on port 8080, while the client should connect to the server using host = 127.0.0.1 and port = 8080. Once connected, the client should send the message "Hello Server!", and the server should display the received message on the screen. Discuss the limitations (disadvantages) of this basic implementation.